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Deep Learning for Clinical Ultrasound Imaging: From Supervised Approaches to Foundation Models

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***Abstract*— Deep learning models have traditionally been developed and trained to perform specific tasks, which limits their generalizability across different domains. Recently, the field has experienced a paradigm shift toward foundation models, wherein AI systems are pre-trained on extensive datasets to support a broad range of tasks across multiple domains. This shift is particularly promising for ultrasound imaging, a relatively low-cost and versatile modality in clinical practice. However, their adoption remains limited due to challenges such as scarce labeled data, domain-specific variability, and the need for user expertise. This review first outlines the workflow and design principles of deep learning-based ultrasound clinical applications, with emphasis on advancements in supervised and self-supervised learning, discriminative models, generative AI models, foundation models, and other miscellaneous models. Recent literature on ultrasound image analysis is then reviewed in detail. Furthermore, deep learning-based ultrasound clinical applications are summarized, and the potential of foundation models to enhance diagnostic performance and expand the scope of ultrasound in clinical settings is highlighted. Finally, key challenges are discussed, and future directions for deep learning in clinical ultrasound are proposed.**

***Index Terms*— Deep learning, foundation models, ultrasound clinical diagnosis, ultrasound imaging, ultrasound image analysis.**

I. INTRODUCTION

Ultrasound is a widely used clinical imaging technique that offers distinct advantages over other imaging modalities such as computed tomography (CT) and magnetic resonance imaging (MRI). This includes its relative low cost, portability for point-of-care use, real-time imaging, and lack of ionizing radiation. Ultrasound can image various human organs (e.g., heart, liver, prostate), superficial structures (e.g., muscle, thyroid), and others (e.g., fetus). Therefore, it is extensively used in clinical practice for human body examinations, most commonly in cardiac imaging, pregnancy assessment, and musculoskeletal tissue evaluation.

Traditional ultrasound clinical diagnosis requires a high level of expertise to collect and interpret data. Operators must properly position the ultrasound probe over the anatomical region of interest, apply an appropriate amount of pressure, and interpret images with a high proficiency to ensure diagnostic accuracy. Consequently, conducting an ultrasound scan is very time-consuming and prone to error. Clinical ultrasound remains largely manual or semi-automatic, which does not fully address the challenges of user dependence. Advances in AI present a promising path to automate ultrasound-based clinical assessment [1], [2]. Ultimately, a fully automated ultrasound system that achieves seamless integration of user guidance, image interpretation, and clinical decision-making could solve many of the challenges associated with operator dependence.

Deep learning (DL) techniques [3] are promising tools for automating clinical ultrasound. In numerous instances, DL-based computer-aided diagnosis (CAD) has proven to have superior generalization ability compared to traditional machine learning approaches [4], and can quickly and accurately achieve end-to-end disease predictions. As a result, DL has been successfully applied to a wide range of ultrasound-based clinical diagnoses and decision-making [5]. For example, DL techniques have demonstrated significant potential in the early detection, screening, and diagnosis of breast cancer [6], [7], [8], [9], [10], [11].

Recent theoretical breakthroughs and practical advances in DL have re-shaped its application in clinical ultrasound, particularly through the development of new learning paradigms and model architectures. To understand how these advances impact ultrasound, it is necessary to explore the strengths and limitations of different DL paradigms, including supervised learning, self-supervised learning, generative models, and foundation models. Each paradigm offers unique contributions to data efficiency, scalability and generalization, which are especially relevant in the ultrasound domain.

Among these, supervised learning has been the most successful as it relies on large-scale annotated datasets. However, data scarcity and the need for annotation have been core bottlenecks. This has led to growing interest in self-supervised learning, which seeks to leverage unlabeled data to improve data efficiency and scalability. Meanwhile, the rise of

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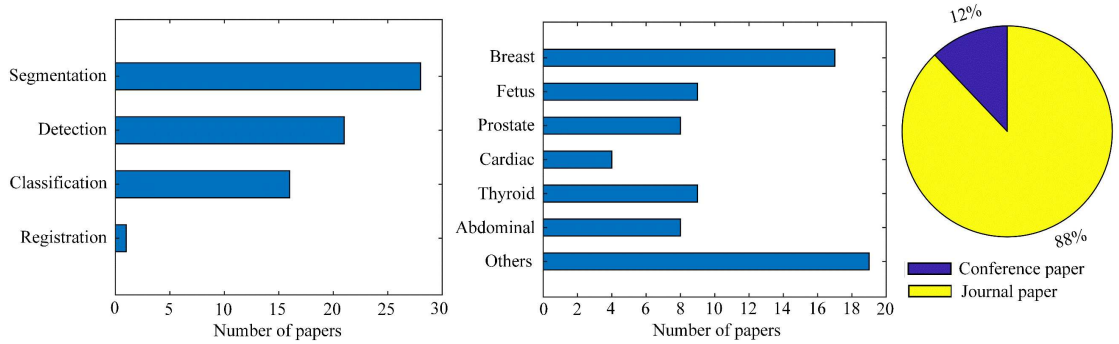

Fig. 1. The statistical breakdown of the literature.

generative AI provides new data sources for training discriminative models. Despite these advances, DL still faces challenges in generalizability and scalability, particularly under real-world clinical variability, data shifts, and heterogeneous ultrasound devices.

Recently, foundation models, which use self-supervised learning, have emerged as a promising direction. These models undergo pre-training on extensive and varied datasets, making them adaptable across a wide spectrum of applications through transfer learning [12]. Notable advancements include: (1) LLMs such as BERT [13], GPT 1.0-3.0 [14], [15], [16], [17] and ChatGPT; (2) large vision models (LVMs) such as the Segment Anything Model (SAM) [18] and large multimodal models (LMMs) such as CLIP [19]. These scalable architectures unify learning across tasks and domains, address key challenges in DL applications, and support the effective utilization of AI models in ultrasound clinical assessment.

In this context, it is essential to review the evolution of DL paradigms to understand their applicability and limitations in ultrasound imaging. Accordingly, this review presents recent breakthroughs in supervised learning, self-supervised learning, generative AI, foundation models, and other miscellaneous models within the ultrasound image analysis domain. Moreover, we explore ongoing challenges and envision potential future opportunities for DL in clinical ultrasound.

We searched the literature using keywords (i.e., "deep learning" AND "ultrasound image" OR "ultrasound" OR "ultrasonography") across major databases (e.g., Google Scholar, IEEE Xplore) for the period 2018-2024. Relevant studies were screened and included from indexed journals and conference proceedings. Papers were excluded if they lacked detailed information about the datasets used. Through this comprehensive effort, we synthesized the collective knowledge of prior work and provided guidelines for future researchers aiming to apply DL to clinical ultrasound applications. The statistical breakdown of the literature is shown in Fig. 1.

Our survey is structured as follows: Section II outlines the workflow and design principles for DL-based clinical ultrasound applications and introduces the main DL techniques and their architectures, with a focus on the state-of-the-art progress of supervised learning, self-supervised learning, generative AI models, foundation models and other miscellaneous models. Section III illustrates the contributions of DL to clinical ultrasound application categories, including breast, fetus, thyroid, prostate, cardiac, abdominal, and others. Section IV summarizes DL-based clinical ultrasound applications, and further emphasizes how foundation models have significant potential for these applications. Sections V and VI discuss challenges in DL-based ultrasound clinical applications and envision future research opportunities.

II. Materials and Methods

A. Workflow for Clinical Ultrasound Applications

Recent advancements in DL have significantly impacted ultrasound imaging, image analysis, and clinical diagnosis. The integration of DL techniques into this field has the potential to enhance accuracy, efficiency, and overall patient outcomes. Fig. 2 outlines the typical workflow for implementing DL-based clinical ultrasound applications. The DL workflow begins with ultrasound data acquisition and progresses toward clinical deployment. The acquired raw ultrasound images are first processed and then fed into DL models (e.g., discriminative models). Meanwhile, generative models can be used to augment the dataset by synthesizing additional images, which enhances the robustness and generalization of the model. Finally, the trained models support various ultrasound clinical applications, thereby facilitating clinical deployment.

The workflow includes the following key steps: first, ultrasound scans are performed for data acquisition according to an experimental protocol. Next, a series of ultrasound image processing approaches are applied to remove artifacts, noise, and distortions, thereby improving the image quality to fit the input of the DL model. Then, the DL model is designed, implemented, evaluated, and adapted for downstream clinical application (e.g., breast, fetus and thyroid). Finally, the AI model is deployed in a clinical setting.

When applying DL techniques to ultrasound clinical diagnosis, several practical considerations must be addressed:

1). Experimental Protocol Design: Experiments are conducted to acquire clinical ultrasound data following specific protocols under various scenarios. Since these tests are often operator-dependent, a high level of operator skill is essential.

2). Ultrasound Data Acquisition: The quality of ultrasound imaging data significantly impacts the performance of DL models. In practice, ultrasound artifacts, distortions, and noise are inevitable. Challenges include low-quality scans, limited

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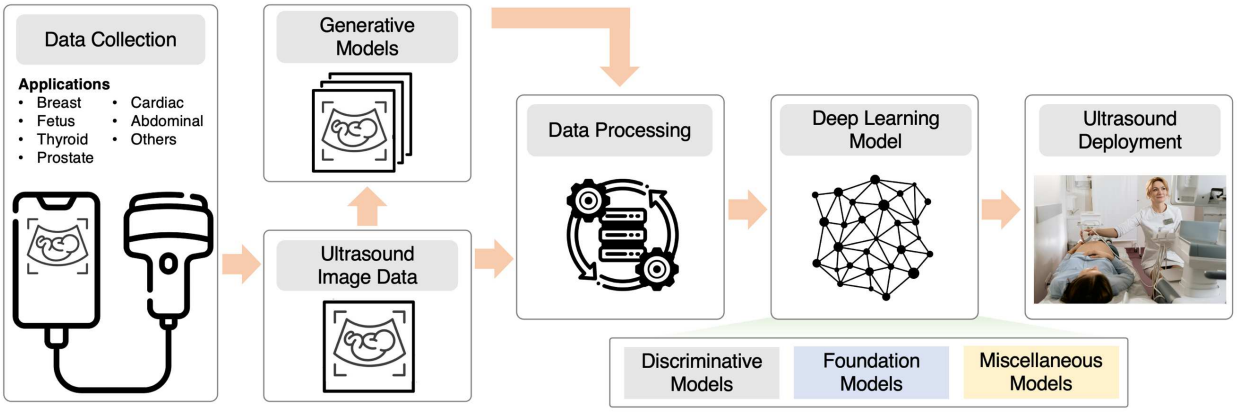

Fig. 2. The overall workflow for DL-based ultrasound clinical applications.

data quality, data heterogeneity, and interpretability issues.

3). Ultrasound Data Preprocessing: Effective preprocessing of ultrasound data is crucial for achieving optimal prediction performance with the DL model. This process involves handling missing data, making necessary annotations, removing noise, distortion, and artifacts, as well as adjusting the image size and pixel resolution.

4). DL Model Selection: The DL model is trained, tested, and validated to identify a robust model that performs well across different datasets. The model can be selected based on the evaluation of its computational performance which includes accuracy, cost, and robustness.

5). AI-based Clinical Deployment: If the AI model performs well, it can be deployed in a real-world setting. This could have significant clinical impact by integrating into healthcare applications such as robotic surgery, virtual assistants, and clinical decision support [21], thereby enhancing its capabilities and potentially transforming ultrasound clinical practices. Before the deployment, the real-time model prediction performance should be carefully addressed. The performance of the deployed model should be continuously monitored, and the model should be updated with new data. Hence, it must be retrained periodically to adapt to changes in data distribution.

B. DL Methodology

DL techniques have been successfully applied to ultrasound image analysis and play a role in automating clinical ultrasound assessment. Recent DL advances are summarized in Fig. 3.

It is evident that DL techniques have evolved significantly in recent years through the following stages: 1) Task-specific supervised learning models, which rely on large-scale annotated datasets, have transitioned to self-supervised learning models that learn representations with limited labels; 2) Foundation models, which can handle a wide range of tasks by being trained on vast amounts of unlabeled data using self-supervised learning, are emerging; 3) Generative AI models, which take inputs such as images, text, and video to generate new content, are transitioning from GANs to diffusion models, as they provide powerful data sources for DL systems in addition to experimental data. In practice, these advancements address current challenges related to DL-based ultrasound image analysis, such as limited raw data and labels, as well as annotation difficulties, and are poised to transform ultrasound clinical diagnosis.

In the following sections, we summarize the architecture of DL models and the recent advancements in supervised learning, self-supervised learning, generative AI models, foundation models, and other miscellaneous models, and then review their application to ultrasound clinical applications.

C. DL Techniques

1) Supervised Learning: DL architectures can be categorized into: convolutional neural networks (CNN) and long short-term memory (LSTM) architectures. U-Net, which leverages skip connections and an encoder-decoder architecture, has become one of the most important architectures in the field of medical image analysis. Meanwhile, since the birth of the fundamental architecture-transformer [22], which is designed based on a self-attention mechanism, representative models such as vision transformer [23] and Swin transformer [24] have emerged for computer vision. These transformer-based architectures have been widely used, and form the basis for foundation models.

Supervised learning is the most mature learning technique and continues to dominate the field of medical image analysis. It is defined as a system that can learn a mapping between input examples (input features x) and a target variable (label pairs y). Consequently, it generally relies on annotated datasets. To date, supervised learning has been successfully applied to various ultrasound clinical tasks (e.g., classification, detection, and segmentation) by training models to map ultrasound images (input) to clinical decisions or labels (output), with emphasis on disease detection, tissue and organ classification, lesion segmentation, and measurement estimation. CNNs and LSTMs are the two most popular architectures [5] for supervised learning. CNNs automatically learn spatial hierarchies of features from 2D or 3D ultrasound images, and are thus suitable for classification, detection, and segmentation. LSTMs, which maintain temporal information, are particularly effective for analyzing ultrasound videos, longitudinal patient data, and motion tracking. Supervised learning methods achieve strong

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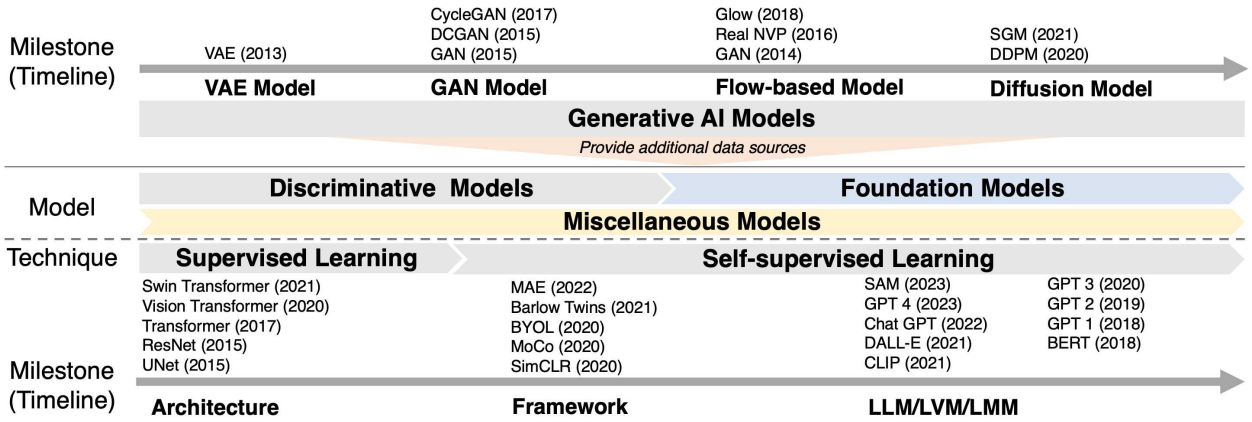

Fig. 3. Overview of the DL paradigm shift toward foundation model.

performance with sufficient labeled data, but high annotation demands and variability across operators and devices limit scalability and generalization.

2). Self-supervised Learning: Self-supervised learning (SSL), which learns representations using unlabeled data, includes contrast learning (e.g., MoCo [25], MAE [26], SimCLR [27], and BYOL [28]) and non-contrast learning (e.g., Barlow Twins [29]). Contrastive methods are well suited for tasks that require fine-grained discrimination and distinction, such as lesion detection, anatomical landmark localization, or small structures segmentation. Non-contrastive methods, which focus more on global structure and contextual information are advantageous for tasks such as tissue characterization, organ classification, or monitoring longitudinal changes like tumor progression. Unlike supervised learning, which uses a retraining strategy, SSL adopts a fine-tuning strategy, allowing it to be adapted to a wide range of downstream tasks. SSL methods can achieve comparable results with much less annotation, making them suitable replacements for supervised learning models under a variety of tasks, such as segmentation [30], [31].

In medical image analysis, there are several challenges including data scarcity, limited training labels, and the complexity of medical data due to privacy protection. Therefore, SSL models have been increasingly used in recent years for medical image analysis, such as image classification [32], as shown in Fig. 3. Similarly, SSL has gradually been applied in the field of ultrasound image analysis for tasks such as image reconstruction [33], accurate liver view classification [34], label-free cardiac segmentation [35], prostate cancer detection [36], endoscopic ultrasound diagnosis [37], contrastive learning for fetal imaging [38], [39], 3D high-resolution ultrasound imaging [40], cardiac ultrasound segmentation, and carotid plaque classification [41], and thyroid ultrasound diagnosis [42]. Overall, SSL that could improve data efficiency and model robustness is particularly promising for ultrasound applications, where labeled data are scarce and anatomical variability is high. However, its performance depends on the choice of pretext task and fine-tuning strategy.

D. DL Models

D.1 Discriminative Models

Discriminative models are a type of DL model designed to classify data by learning the decision boundary between different classes. They primarily focus on distinguishing between categories based on feature patterns. These models are commonly used in classification tasks where the goal is to map input features directly to a specific label.

Discriminative models play a dominant role in ultrasound clinical applications such as disease classification and organ segmentation. While effective with sufficient labeled data, their reliance on task-specific training and sensitivity to variability across sites, devices, and operators can limit generalization and clinical deployment.

D.2 Generative Models

Discriminative models typically require large-scale datasets. However, scarcity of ultrasound clinical data continues to limit their application. To address this challenge, generative models have been developed to augment ultrasound image datasets and to address data scarcity challenges in analysis. Two primary approaches exist: 1) ultrasound image generation based on other modalities such as CT or MRI; and 2) ultrasound data synthesis from limited ultrasound image datasets. Currently, there are four representative generative models: GANs, diffusion models, variational autoencoders (VAEs) and flow-based generative models.

Prior to 2020, few studies addressed the use of generative AI models for ultrasound data augmentation. More recently, the number of related publications has gradually increased, and it is evident that current mainstream generative AI models used for augmenting ultrasound datasets are GANs and their variants. These models have been widely employed to generate synthetic ultrasound image data, including for breast [43], [44], musculoskeletal [45], intravascular [46], and others [47], to facilitate DL model training. More recently, diffusion models have also been employed to synthesize ultrasound data, such as musculoskeletal ultrasound images [48].

Generative Adversarial Networks (GANs): GANs learn to generate new data that have the same statistics as the training

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set, with numerous research studies demonstrating its effectiveness on small-scale datasets. GAN variants such as DCGAN [49], CycleGAN [50] and StyleGAN [51] have been proposed and widely used in recent years. They are popular in the field of medical image analysis for generating synthetic medical data for data augmentation. GANs excel at high-fidelity ultrasound image synthesis. Particularly, conditional GANs (e.g., CycleGAN) are powerful for translating low-quality ultrasound images into high-quality ones or translating images between different modalities (e.g., MRI to ultrasound). In the literature, GANs have been used for low-level vision such as ultrasound video high-quality reconstruction [52] and ultrasound image super-resolution [53]. They are used to synthesize various ultrasound images such as fetal ultrasound brain plane [54]. GANs could serve for various downstream applications including thyroid nodules identification [55], cardiac ultrasound image segmentation [56], and breast and mammography mass classification [57]. For example, the musculoskeletal ultrasound images are generated by CycleGAN model [45].

Diffusion Models: These models generate a probability distribution of a given dataset by learning a two-stage diffusion process: a forward diffusion stage and a reverse diffusion stage. There is a growing interest in the application of diffusion models to medical image generation. Variants such as the noise-conditioned score network [58] and denoising diffusion probabilistic models [59] have gradually been used in image synthesis. Diffusion models can generate exceptionally high-quality and diverse ultrasound images. Their iterative denoising process allows for fine-grained control and produces ultrasound images with very realistic details, which is valuable for medical ultrasound applications. In the literature, diffusion models have been used to synthesize high-quality musculoskeletal ultrasound images [48]. Additionally, research has demonstrated that diffusion models can outperform GANs in image synthesis [60].

VAEs: VAEs provide a probabilistic framework for describing an observation in latent space [61]. It utilizes an architecture that comprises an encoder-decoder structure. VAEs are excellent for learning a smooth, continuous latent space. This allows for meaningful interpolation, where you can move between two different ultrasound images in the latent space to generate a smooth transition of synthetic ultrasound images. This is valuable not only for understanding disease progression or anatomical variations, but also for tasks such as anomaly detection, interpolation, and controllable synthesis. VAEs have shown promise in new content generation [62], anomalies detection [63], and noise removal [64]. Additionally, they can also be used for ultrasound image inpainting [65].

Flow-based Generative Models: These models are constructed using a sequence of invertible transformations. Typical models that employ normalizing flows are NICE [66], Real NVP [67] and Glow [68]. Flow-based generative models could accurately compute the likelihood of a generated or real image. This is valuable for model evaluation and for applications such as anomaly detection, where outliers can be identified by their low likelihood. They have successfully been used for anomaly detection in medical images [69]. Thus far, there are very few papers exploring their application to ultrasound dataset augmentation.

Overall, generative models are powerful tools that can increase data diversity and enhance image quality, but their generated synthetic images must be validated carefully to avoid introducing artifacts or distribution shifts that could reduce clinical performance.

D.3 Foundation Models

Most DL models trained based on limited datasets are task specific, which often lack flexibility and adaptability. In contrast, foundation models are trained on broad datasets using self-supervised learning, and thus can be adapted to a wide range of downstream tasks [12]. LLMs refer to a type of AI model which are trained on vast amounts of text data and can perform a variety of language-related tasks, including text generation, translation, summarization, and more. Typical examples include GPT 1.0-3.0 and ChatGPT, which have recently achieved notable success. While LLMs originated from NLP, they have recently transformed the field of computer vision [70]. Similarly, LVMs refer to a type of AI model that is trained on extensive datasets containing images, videos, or other visual data, enabling them to perform various vision-related tasks such as classification, object detection, image generation, segmentation, and visual question answering. A recent milestone is the release of SAM, which is a large vision model. SAM and its variants (e.g., MedSAM) have been increasingly used for medical image segmentation since their release. Overall, foundation models are reshaping the design, development and deployment of DL models and hold significant promise for impact in digital health, medical robotics, medical imaging, and health informatics [71].

LLMs: LLMs are a class of language models designed based on the transformer architecture and trained with a wide range of large-scale textual datasets to perform a series of NLP tasks. Typical models such as GPT 1.0-3.0 and ChatGPT, released by OpenAI, as well as PaLM [72] and LLAMA [73] have garnered significant attention in recent years. LLMs can assist in making intelligent diagnoses by leveraging their ability to process natural language. They have strong potential for improving the efficiency and effectiveness of clinical decision making [74]. Although research on the applications of LLMs to ultrasound clinical diagnosis is still in the early stage, LLMs hold promising potential for future clinical applications.

LVMs: LVMs are a class of vision models designed based on the transformer architecture and trained with a wide range of large-scale image datasets to perform a series of vision tasks. A representative LVM model is the SAM model (https://segment-anything.com/) [18]. This model was introduced as the first visual foundation model for natural image segmentation and has demonstrated strong zero-shot capabilities.

SAM consists of an image encoder, a prompt encoder, and a mask decoder. Its variants (e.g., SAM3D [75]) have been developed for various applications. An experimental study

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evaluated SAM across different medical image datasets [76], revealing that SAM does not perform as well as U-Net, due to significant differences between natural and medical images. Several SAM variants such as MedSAM [77] and AutoSAM [78] have been proposed to improve segmentation accuracy. Recently, a universal ultrasound foundation model (USFM) [79] was introduced.

LMMs: LMMs, also called multimodal foundation models, leverage image, text, audio, video, and other sensory modalities, inheriting all the properties of LLMs/LVMs. Typical LMMs include CLIP [19], DALL-E [80] and GPT-4 [81]. Inspired by CLIP, an ultrasound LLMs-EchoCLIP [82], which leverages 1,032,975 cardiac ultrasound videos and corresponding expert interpretations, was proposed for echocardiogram interpretation.

So far, foundation models in clinical ultrasound have shown promising results. For example, the superiority of foundation model SAMUS [83] over task-specific methods has been demonstrated for ultrasound image segmentation. Robust performance could be obtained with the USFM [79] by using only 20% annotation for image segmentation, classification, and enhancement tasks. In addition, an area under the curve (AUC) of 0.77-0.86 could be achieved with the EchoCLIP [82] for predicting unique patient and clinical transitions in echocardiography studies. Despite these powerful early results, their widespread clinical use remains limited. Key challenges persist, including the necessity for large-scale, diverse ultrasound datasets, the development of effective fine-tuning strategies for different clinical tasks, and constraints related to real-time deployment on resource-constrained devices.

D.4 Miscellaneous Models

Explainable AI: Transparency and interpretability of DL models are essential in healthcare applications. Techniques such as feature importance analysis and visualization are utilized to understand how the model makes predictions. Task-specific models generally have clear input and output domains, making the behavior easily interpretable to a user. Despite advances in interpretable AI, confounding factors from image and video artifacts may still affect model predictions. To address this challenge, interpretation techniques such as CAM [84] and Grad-CAM [85] can be used, with their performance thoroughly benchmarked through iterative discussions with the clinical collaborators. The Grad-CAM method computes the importance of each input dimension across multi-site images to identify interpretable factors for prediction. Specifically, Grad-CAM computes the gradient of the trained model with respect to the input data, which is then processed into heatmaps to highlight regions that significantly contribute to the final prediction [86].

In the literature, explainable AI has many applications. MediMLP which uses Grad-CAM is applied to identify critical variables for lung cancer postoperative complication prediction [87]; CAM is used to show how B-lines that enter the imaging plane are well-indicated [88]; The heatmaps produced by Grad-CAM and the bounding boxes obtained by thresholding are reported [86]. Explainable AI has also been applied to ultrasound for accelerating the detection of lung pathology [89]. By utilizing CAM/Grad-CAM to generate heatmaps, clinicians can understand crucial regions of ultrasound images that most contribute to disorder classification or prediction, and easily identify medically relevant regions such as lesions. Although these methods improve transparency and clinician trust, their spatial accuracy, robustness, and clinical interpretability remain limited in ultrasound imaging.

Human-in-loop Learning and Active Learning: The human-in-loop learning model integrates active learning and annotation for human-centered AI [90]. Active learning with stochastic batches is efficient for medical image segmentation [91], [92]. DL models rely on large-scale annotated datasets, but manual data annotation is expensive, tedious, and time-consuming. To minimize manual annotation, active learning focuses on selecting the most informative samples from the unlabeled set for annotation, which are then added to the labeled training set. This approach allows for the selection of optimal data to enhance model performance. However, in the literature, it has received limited attention in ultrasound clinical diagnosis. For example, pseudo-label divergence-based active learning was introduced to segment carotid intima-media in ultrasound images [93]. These approaches offer clear advantages in reducing annotation burdens and improving generalization in data-scarce settings; however, reliance on expert input, complex workflows, and limited scalability continue to hinder clinical adoption.

III. APPLICATIONS

A. Breast

An overview of published papers on DL-based breast ultrasound image analysis is shown in Table I.

Application: DL techniques have demonstrated excellent performance in segmenting breast lesions [95], [96], [100], classifying breast cancer [11], detecting breast cancer [103], and performing related tasks (e.g., localization and classification of masses in breast ultrasound images [94]). In addition, they are commonly applied in image preprocessing, such as denoising, to enhance breast ultrasound image quality [102].

Methodology: To date, U-Net [104] and its variants, such as Attention-U-Net [105], have dominated the field of breast ultrasound image segmentation due to their high performance. CNNs (e.g., ResNet) have been extensively used in breast ultrasound image classification. Following the introduction of the transformer architecture, hybrid models such as HCTNet [100], which integrate CNNs and transformer components, have been developed to improve overall accuracy.

Perspective: DL for breast ultrasound has progressed from CNNs to hybrid models. However, as shown in Table I, models rely on small datasets (around 1,000 images) and often lack external or prospective evaluations. Clinical deployment also faces barriers such as domain shift caused by ultrasound device and acquisition parameter variability, noisy ground truth labels, limited robustness of model prediction and segmentation, and mismatches between static-frame models and cine-based

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TABLE I
OVERVIEW OF PUBLISHED PAPERS OF DL-BASED BREAST ULTRASOUND IMAGE ANALYSIS

References	Year	Model	Datasets	Description
Shin et al.[94]	2018	Fast R-CNN	10 strongly annotated images, 800 strongly annotated images	localization and classification of masses in breast ultrasound images
Vakanski et al. [95]	2020	Attention, U-Net	510 BUS images	breast tumor segmentation
Chen et al. [9]	2021	ResNet	Breast-CEUS Dataset	breast cancer diagnosis
Chen et al. [96]	2022	AAU-Net attention U-Net	3 BUS datasets (780: 163: 42)	breast lesions segmentation
Jabeen et al.[10]	2022	DarkNet-53	Normal (133 images), Malignant (210 images), Benign (437 images)	breast cancer classification
Qu et al. [97]	2022	VGG16, ViT	two BUS datasets (974 samples and 163 samples)	benign and malignant classification of breast ultrasound images
Li et al. [98]	2022	ResNet	780 samples (133 normal samples, 487 benign samples, and 210 malignant samples)	breast tumor lesion detection
Gheflati and Rivaz [99]	2022	Vision Transformer	Two datasets (780 images and 163 images)	breast ultrasound images classification
He et al. [100]	2023	HCTNet: A hybrid CNN-transformer	3 breast ultrasound image datasets 1. BUSI 2. BUS 3. Dataset B	breast ultrasound image segmentation
Atrey et al.[11]	2023	CNN+LSTM	86 images consisting of 43 mammogram and 43 ultrasound images	breast cancer classification
Deb and Jha. [101]	2023	VGG-16, DenseNet, Xception, Inception	Breast Ultrasound Image dataset, consists of 780 images which comprises 210 malignant images, 437 benign images, and 133 normal images	breast ultrasound classification
Vimala et al. [102]	2023	Logical-Pool Recurrent Neural Network (LPRNN)	INbreast dataset (412 images), CBIS-DDSM dataset (1000 images)	image noise removal
Sahu et al. [103]	2024	AlexNet, ResNet, and MobileNetV2	BUSI (contains 780 images), BUS2(contains 250 images)	breast cancer detection

clinical workflows.

B. Fetus

An overview of published papers on DL-based fetal ultrasound image analysis is shown in Table II.

Application: DL techniques have demonstrated strong performance in various fetal classification tasks (e.g., fetal brain [108], fetal standard plane [111], [113]), fetal anatomical segmentation [113] and fetal head detection [106]. In addition, DL-based image reconstruction has been applied to fetal ultrasound imaging [107].

Methodology: U-Net [104] has been successfully used for fetal head detection. Additionally, vision transformer-based architectures such as COMFormer [109] have been introduced to classify maternal-fetal and brain anatomy. GANs combined with CAMs have been used to synthesize realistic fetal brain ultrasound images [110]. Self-supervised learning, such as SimCLR, have also been applied for detection and

TABLE II
OVERVIEW OF PUBLISHED PAPERS OF DL-BASED FETAL ULTRASOUND IMAGE ANALYSIS

References	Year	Model	Datasets	Description
van den Heuvel et al. [106]	2019	UNet	183 pregnant women in the hospital	fetal head detection
Gomez et al. [107]	2019	VAE	Synthetic (8 images) and real datasets (2 patients)	image reconstruction applied to fetus ultrasound imaging
Xie et al. [108]	2020	CNN	15,372 normal and 14,047 abnormal fetal brain images in standard axial planes	fetal brain classification
Sarker et al. [109]	2023	COMFormer	BCNatal consisting of 12,400 images from 1,792 subjects.	classification of maternal-fetal and brain anatomy
Lasala et al. [110]	2023	GAN, CAM	12,400 images from 1,792 patients	fetal brain image synthesis
Migliorell et al. [111]	2024	SimClr, Siamese	12,400 images from 1,792 patients	fetal ultrasound standard plane classification
Fiorentino et al. [112]	2025	SimClr, kNN	two multi-centric datasets	fetal standard plane detection
Ambsdorf et al. [113]	2025	UltraDINO	A fetal ultrasound dataset of 2M images	standard plane classification, anatomical segmentation

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TABLE III
OVERVIEW OF PUBLISHED PAPERS OF DL-BASED THYROID ULTRASOUND IMAGE ANALYSIS

References	Year	Model	Datasets	Description
Li et al. [114]	2018	CS Faster R-CNN (Improved Faster R-CNN)	VOC2007 database	thyroid papillary cancer detection
Wang et al. [8]	2020	VGG16	3120 images (1841 benign nodules and 1393 malignant nodules) from 1040 cases	thyroid nodules classification
Hu et al. [115]	2023	DSC-Net: vision transformer, ResNet	UDIAT, BUSIS, and LUSI	thyroid nodule segmentation
Chen et al.[116]	2023	swin-transformer encoder, UNet	CEUS dataset thyroid nodule dataset 163 cases including 87 malignant nodules and 76 benign nodules	thyroid nodule segmentation
Lin et al. [83]	2023	SAMUS	US30K with 30,106 images with 68,570 masks	thyroid nodule, thyroid gland, left ventricle, myocardium, and left atrium segmentations
Ye et al. [117]	2023	Swin Transformer	3,853 US images from 3,000 cases.	thyroid nodule detection

classification tasks [111], [112]. Recently, the foundation model UltraDINO has been employed for classification and segmentation tasks [113].

Perspective: DL for fetal ultrasound has progressed toward self-supervised learning and foundation models. However, many models are trained and tested on datasets such as BCNatal collected from limited centers [108], [109], [110], [111]. Barriers to clinical deployment include high interclass variability [111], multi-expert image annotation difficulty, the computational time for AI diagnosis [108], and lack of a systematic evaluation workflows.

C. Thyroid

An overview of published papers on DL-based thyroid ultrasound image analysis is shown in Table III.

Application: Most DL-based studies primarily focus on segmentation, detection, and classification of thyroid nodules.

Methodology: Traditional mainstream DL-based object detectors, such as Faster R-CNN [118], have been employed to detect papillary thyroid carcinoma in ultrasound images [114]. Vision transformers, such as the Swin Transformer, have been widely applied to thyroid nodule segmentation [115], [116], while CNN models are used for nodule classification [8]. Furthermore, foundation models such as SAMUS [83], which is derived from the SAM architecture, have recently been used for ultrasound segmentation.

Perspective: DL for thyroid ultrasound has progressed from CNNs to Transformer models that better extract nodule features [115], [117]. However, validation gaps persist due to small datasets (Table III), data bias, and limited external validation [116]. Barriers to clinical deployment include high acquisition variability, small lesions and class imbalance, low accuracy for challenging cases [8] and limited generalization capacity of DL models. Moreover, foundation models such as SAMUS [83] are computationally demanding, which poses a challenge for real-time use in clinical settings.

D. Prostate

An overview of published papers on DL-based prostate ultrasound image analysis is shown in Table IV.

Application: The majority of existing work focuses on the segmentation [119], [122], [123], [124] and detection [120], [121] of prostate cancer using ultrasound and histopathology images. These tasks are critical for DL-based CAD systems.

Methodology: CNNs and RNNS are widely employed for prostate cancer detection, while U-Net and its variants are the predominant architectures used for prostate segmentation. In addition, a foundation model, ProstNFound [125] has recently been proposed for prostate cancer detection.

Perspective: DL for prostate ultrasound has evolved from

TABLE IV
OVERVIEW OF PUBLISHED PAPERS OF DL-BASED PROSTATE ULTRASOUND IMAGE ANALYSIS

References	Year	Model	Datasets	Description
Anas et al. [119]	2018	ResUnet+, ConvGRU	2,238 labeled transrectal ultrasound images, with an additional 637 and 1,017 unseen images used for validation and testing	prostate segmentation
Feng et al.[120]	2018	3D CNN	The CEUS Sample Sets: 21,841 (target agent) and 25,738 (non-target agent)	prostate cancer detection.
Azizi et al.[121]	2018	RNN	data from 255 prostate biopsy cores of 157 patients	prostate cancer detection
Karimi et al. [122]	2019	U-Net	B-mode axial TRUS images of 675 brachytherapy patients.	prostate clinical target volume segmentation
Orlando et al. [123]	2020	U-Net	246 3D TRUS images consisted of 104 end-fire and 142 side-fire 3D TRUS images	prostate segmentation
Vesal et al. [124]	2022	dilated Residual UNet	three large datasets encompassing 2067 subjects	prostate segmentation
Wilson et al. [125]	2024	ProstNFound	a multi-center micro-ultrasound dataset with 693 patients	prostate cancer detection

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TABLE V
OVERVIEW OF PUBLISHED PAPERS OF THE DL-BASED CARDIAC ULTRASOUND IMAGE ANALYSIS

References	Year	Model	Datasets	Description
Smistad et al. [126]	2020	UNet	500 patients from an outpatient clinic were used to train deep neural networks, while a separate data set of 100 patients from another clinic was used for evaluation.	automatic ejection fraction and foreshortening detection
Komatsu et al. [127]	2021	UNet	247 normal fetal ultrasound videos	cardiac structural abnormalities detection
Vukadinovic et al. [128]	2024	EchoPrime	A dataset of 12,124,168 2D and color Doppler echocardiography videos	echocardiography interpretation
Deng et al.[129]	2024	MemSAM	CAMUS (500 cases, including 2D apical two-chamber and apical four-chamber view video) and EchoNet-Dynamic (10030 2D apical two-chamber view videos)	echocardiography segmentation
Kim et al. [130]	2024	EchoFM	over 20 million echocardiographic images	echocardiogram analysis
Kim et al. [131]	2025	MediViSTA	CAMUS, EchoNet-Dynamic, and a multi-center internal dataset	echocardiographical video segmentation
Chao et al. [132]	2025	SAM	CAMUS, EchoNet-Dynamic, Mayo Clinic data set	left ventricular segmentation

CNN/RNN to foundation models. However, most studies lack multi-center external validation, prospective trials, and systematic evaluation of inter- and intra-observer variability [119], [123]. Clinical deployment is hindered by model instability under different conditions (e.g., rapid ultrasound probe movement [119]), severe domain shifts, subtle lesion contrast, and noisy biopsy-derived labels. Although the foundation model ProstNFound [125] shows its promising potential for clinical application, its prospective validation and clinical translation still worth further exploration.

E. Cardiac

An overview of published studies on DL-based cardiac ultrasound image analysis is shown in Table V.

Application: A significant body of work [129], [131], [132] has focused on echocardiography segmentation. DL techniques have also been applied to detection tasks, such as identifying cardiac structural abnormalities [127]. In addition, DL has been utilized for tasks including automatic ejection fraction and foreshortening detection [126], which are essential components of DL-based computer-aided echocardiography diagnosis.

Methodology: Conventional UNet models remain widely used in the cardiac ultrasound applications. More recently, several foundation models, including EchoPrime [128], MemSAM [129], EchoFM [130], and MediviSTA [131], have been developed for echocardiographic video segmentation and interpretation. These models represent early, notable examples of applying foundation model paradigms to ultrasound image analysis. On the other hand, a fine-tuned SAM has

TABLE VI
OVERVIEW OF PUBLISHED PAPERS OF THE DL-BASED ABDOMINAL ULTRASOUND IMAGE ANALYSIS

References	Year	Model	Datasets	Description
Sun et al.[7]	2019	ResNet-50	462 liver tissue slide images, containing 79 normal liver histopathological images and 383 liver hepatocellular carcinoma images	classification of liver cancer histopathology images
Yin et al. [133]	2020	VGG16, Deeplab	2,238 labeled transrectal ultrasound images, with an additional 637 and 1,017 unseen images used for validation and testing, respectively	automatic kidney segmentation
Reddy et al. [134]	2021	CNNs	A total of 1906 B-mode ultrasound images are collected from 983 consecutive clinical abdominal ultrasound studies	classification of abdominal organs (kidney, liver, pancreas, spleen, and urinary bladder)
Chou et al. [135]	2021	CNNs	21,855 images from 2,070 patients classified as normal (N = 11,307), mild (N = 4,467), moderate (N = 3,155),	severity of fatty liver
Xi et al. [136]	2021	ResNet50	a training set 330,000 images and a test set of 79 images.	differentiation of benign and malignant solid liver lesions
Wei et al.[137]	2021	U-Net	1035 clinical images from 52 patients	2D ultrasound and 3D CT/MR image registration in liver tumor ablation
Wang et al. [138]	2023	U-Net	675 rectus abdominis ultrasound images	rectus abdominis segmentation and distance measurement
Kanauchi et al. [139]	2023	YOLOv5, UNet++	2664 annotated abdominal images from 1444 patients	Automatic detection and measurement of renal cysts

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demonstrated excellent segmentation performance and strong generalization capacity [132].

Perspective: DL for cardiac ultrasound has advanced from UNet to foundation models. However, gaps in validation are small datasets which encompass incomplete data from limited institutions [127] or a narrow coverage of cardiac conditions [130], dataset bias, particularly those including rare or atypical cardiac events [130], and video segmentation performance instability [129]. Moreover, clinical deployment is challenged by high computational cost, arrhythmias and varying frame rates that complicate temporal aggregation, and limited flexibility and accuracy of the model owe to its dependency on prior frames [131].

F. Abdominal

An overview of published papers on DL-based abdominal ultrasound image analysis is shown in Table VI.

Application: Most DL research on abdominal organs focuses on the liver. Relevant studies include fatty liver assessment and severity classification [135], as well as liver cancer classification [7]. DL-based kidney segmentation has also been explored [133]. In addition, researchers have investigated DL methods for detecting various abdominal organs (e.g., kidney, liver, pancreas, spleen, and urinary bladder) in ultrasound images [134].

Methodology: CNNs have been widely employed for classification tasks. For organ segmentation, architectures such as Deeplab and UNet++ have been used. Furthermore, traditional DL-based object detectors, such as YOLO [140], have been employed for the automatic detection of the renal cysts [139].

Perspective: DL for abdominal ultrasound primarily relies on CNNs. Key validation obstacles include the lack of external prospective cohorts [135], [139], small datasets [138], data bias [135], inter-subject variability that occurs in abdominal organs [133], and inherent inter- and intra-organ variability [136]. In addition, clinical deployment includes the accuracy and robustness of DL models (e.g., some of them depend on presence of vessels [137]), difficulty of imaging organs that are behind bone and air, and noisy or surrogate ground truths.

G. Others

Application: DL has a variety of applications in ultrasound image analysis. For example, it is widely used for musculoskeletal segmentation (e.g., myotendinous junction [141], transverse musculoskeletal [142], muscle cross-sectional area [143]), muscle boundary [144] and architecture [145] analysis. In addition, considerable attention has been devoted to DL-based biometric measurements, such as bladder volume estimation [149] and plaque area measurement [150]. DL techniques have also recently demonstrated excellent performance in other applications, including intravascular segmentation [146], [147], median nerve segmentation [148], abnormality in cerebral blood flow velocity detection [151], detection of inferior vena cava [152], respiratory motion estimation [153], carotid intima-media thickness measurement [154], B-line detection and localization [155], and B-line identification and quantification [156]. Furthermore, researchers have explored DL applications for detecting COVID-19 in lung ultrasound images [157].

Methodology: The U-Net model and its variants (e.g., U-Net++, Attention U-Net) have been widely applied for segmentation tasks, while CNN architectures (e.g., VGG, ResNet) have been extensively used for classification.

Additionally, CAM techniques are used to visualize and highlight specific regions responsible for detections [88]. Grad-CAM is employed to dynamically visualize the localization of COVID-19 markers [86]. However, to date, few studies have investigated foundation models in this domain.

Perspective: DL for other ultrasound applications has progressed toward explainable CNN models. Gaps in validation include limited external multi-site validation [144], [151], [154], extreme data heterogeneity caused by anatomical variability in ultrasound images [148], inconsistent ground truth and data bias [142], [150]. In addition, clinical barriers include domain shifts, limited generalization, robustness and consistency of DL model, real-time constraints, and hardware limitations, all of which impede clinical deployment.

IV. DISCUSSION

A. DL-based Ultrasound Clinical Applications

Lesion detection, segmentation, and classification are central to automated ultrasound diagnosis. Practical clinical problems directly drive the ultrasound system application and the relevant research. Current research topics of interest include breast lesion detection, segmentation, and classification, which is consistent with findings from a previous review [5]. The most common DL architectures used in these clinical applications are ResNet and U-Net, all of which mainly use supervised learning methods. Moreover, model design is task-specific. In contrast, foundation models offer broader adaptability across multiple tasks, which may help reduce the need for separate specialized models and support more scalable clinical deployment.

B. DL Architectures for Ultrasound Application

Advancements in DL-based computer vision techniques have significantly influenced ultrasound image analysis and clinical diagnosis in recent years. Currently, the most widely used neural network architectures include ResNet, VGG, U-Net and Faster RCNN, which are consistent with the investigation of a previous review [158]. In addition, various transformer-based architectures, such as vision transformer and Swin transformer, have been increasingly important for ultrasound image segmentation and classification. The growing number of relevant publications reflects this trend, which is aligned with prior statistics of the applications of transformers in medical image analysis [159]. Recently, Transformer architectures have been widely leveraged for foundation models, and have shown promising results in clinical applications such as cardiac function assessment [82] and prostate cancer detection [125].

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C. DL Paradigm Shift: Toward Foundation Models

The shift from traditional supervised learning to foundation models in DL has the potential to revolutionize ultrasound clinical diagnosis. While supervised learning models have long dominated the field of medical image analysis, there is a growing trend toward self-supervised learning models, which are increasingly adopted for ultrasound clinical assessment [34], [160], [161]. Generative AI models also play a crucial role by augmenting ultrasound datasets to train DL systems for improved model robustness and accuracy. Foundation models are emerging as powerful tools capable of performing fundamental tasks, such as segmentation [79], [82], [83], with their inherent adaptability allowing for fine-tuning in specific ultrasound applications. Below, we outline several key areas where foundation models have the potential for substantial impact in ultrasound clinical diagnosis:

1). Flexible Interactions and Q&A: Foundation models could facilitate more intuitive and adaptable interactions between users and ultrasound devices [162]. Rather than relying solely on rigid workflows or pre-set commands, operators and clinicians could engage with the system using natural language questions and commands. By enabling interactive Q&A capabilities, foundation models are considered self-supervised encoders that can create distinctive representations of ultrasound clinical data, which could promote a more collaborative approach to patient care. In point-of-care settings, where quick decision-making is often essential, clinicians could ask specific questions such as, “What do these measurements indicate for this patient?” This would allow the clinician to receive immediate, contextually relevant feedback without having to sift through manuals or training materials.

2). Multimodal Inputs: Foundation models can synthesize information from multiple sources, such as laboratory results, patient history, and other imaging modalities like MRI or CT scans [12]. By combining these diverse inputs, the model could offer a more comprehensive view of a patient's condition, enabling more personalized and accurate assessments. The multimodal approach could, for instance, integrate ultrasound data with patient demographics and previous medical histories to improve the ability to distinguish between benign and malignant lesions.

3). Dynamic Task Specification: Foundation models have the ability to adapt dynamically to different tasks based on user input [162]. In ultrasound, this means the system could seamlessly switch between measurement or diagnostic tasks without extensive reconfiguration. For example, a user could ask the model to “measure fetal head circumference” and then quickly shift to “assess cardiac function,” with the model adjusting its processing and output accordingly. This adaptability could streamline workflows and reduce cognitive load, allowing clinicians to focus more on patient care rather than navigating complex systems.

4). Automated Measurements: Foundation models excel in automating time-consuming and complex measurements. They can, for example, automatically calculate ejection fraction from cardiac ultrasound images or determine the size and volume of organs or lesions. This automation could not only save time, but also reduce measurement variability, Furthermore, by minimizing human errors, automated measurements can foster greater confidence in the results, allowing clinicians to focus on interpretation and patient care rather than on the technical aspects of data acquisition.

5). Improved Diagnosis: By leveraging vast amounts of training data, foundation models could detect subtle patterns and anomalies that may be missed by human observers or less advanced algorithms [79]. For example, they could identify early signs of diseases like liver fibrosis or detect microvascular complications in diabetic patients with greater precision. Additionally, their ability to continually learn from new data ensures ongoing improvements in diagnostic performance.

Despite their potential, foundation models face several critical limitations. One major limitation is hallucination [163]. Additionally, foundation models are highly dependent on the quality and diversity of clinical training data, making them susceptible to bias [164]. Other limitations include model interpretability and transparency, high computational demands, lack of true reasoning capabilities, and common-sense knowledge. Although medical foundation models [128], [129], [130], [131] have been proposed and applied to ultrasound, their progress has been slower compared to other imaging modalities. Challenges include the real-time and operator-dependent nature of ultrasound, high variability in ultrasound images, lack of standardization in ultrasound clinical data, scarcity of large-scale annotated ultrasound datasets, and hardware constraints. These factors impact the clinical application of foundation models in ultrasound. Addressing these issues will require advancements in domain adaptation, efficient model deployment, and improved dataset curation to enhance the application of foundation models in ultrasound. Recently, a notable effort is by Jiao et al. [79], who employed a masked image modeling (MIM)-based spatial-frequency dual masking method to address the database insufficiency, low quality and ineffective feature representation.

V. Challenges

A. Equipment, Experiment, and Data Acquisition

High-quality ultrasound data generation relies heavily on ultrasound hardware design. Recent advancements in 3D to 6D ultrasound systems [165], [166], [167] have significantly improved imaging quality. However, artifacts, noise, and distortion remain prevalent. In addition, the quality of ultrasound data acquisition is heavily influenced by clinical procedures, which are operator-dependent and can inevitably lead to data variation and heterogeneity. The proficiency of the operator directly impacts the quality of the ultrasound images obtained. Therefore, standardized protocols and continuous training programs for healthcare professionals are essential.

B. Dataset and Benchmark

A large, well-designed, well-labelled, diverse, and multi-institutional dataset may be even more important than model

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optimization [168]. Currently, there are no public large-scale ultrasound datasets comparable to benchmark datasets such as ImageNet [169]. The application of foundation models in clinical ultrasound is hindered by the lack of large-scale, standardized ultrasound datasets. While there are some publicly available datasets, such as POCUS [170], CAMUS [171] and EchoNet [172], they have limitations including lack of multi-modal ultrasound data, limited scale and diversity, and device-specific data collection, all of which affect their suitability for training foundation models. In addition, many researchers rely on private clinical datasets collected from hospitals, which makes it difficult to replicate results or establish benchmarks in ultrasound image analysis.

C. Model Interpretation, Transparency, and Deployment

Model interpretation and transparency are crucial for fostering trust and adoption of DL models in clinical practice. Explainable AI techniques can visualize salient image regions that contribute to model predictions. However, many existing models lack adequate interpretation and transparency. Furthermore, when deployed in real-world clinical environments, the distributional shifts in input data can significantly degrade model performance, as many DL systems often struggle to generalize across diverse patient populations and imaging conditions [173].

VI. FUTURE DIRECTIONS

Ultrasound clinical assessment, shaped by new clinical requirements, has been significantly affected by advancements in ultrasound systems and AI models. With ongoing improvements in sensing technologies, DL models, and evolving clinical needs, several paths forward can be identified:

A. Establish Real-time and Accurate Ultrasound Image-based Feedback

Recent advancements in DL have significantly enhanced our ability to understand underlying patterns from expert demonstrations, which enables the implementation of real-time ultrasound imaging feedback. This progress has facilitated the development of intelligent Robotic Ultrasound Systems (RUSS) [174] and innovative training methodologies for ultrasound practitioners. Given the recent advancements in this field, RUSS holds promise to transform ultrasound-based medical interventions, clinical screenings, and diagnostic workflows. Future efforts should focus on designing novel hardware for human-in-the-loop systems capable of regulating contact force and assisting operators in collecting high-quality scan data.

B. Leverage Generative AI Model for Powerful Data Sources

Generative AI models offer a powerful solution for creating robust data sources. Traditional data augmentation techniques often do not align with the underlying physics of ultrasound and, if applied incorrectly, can produce unrealistic images. Generative AI models can generate reliable synthetic data, which can address data scarcity challenges and annotation difficulties, while also mitigating privacy and security concerns associated with sensitive health data. For instance, recent literature highlights the use of diffusion models to synthesize musculoskeletal ultrasound images [48], and GANs to generate ultrasound images [45]. In the future, it is critical to explore further applications of generative AI models in DL tasks, particularly in conjunction with foundation models.

C. Extend Multimodal Applications of Foundation Models

Vision-language foundation models have demonstrated promising results in domains such as computational pathology [175], [176]. For example, SAM [18] has exhibited automatic segmentation capabilities for natural images, and have showcased robust zero-shot performance compared to traditional DL methods. This has paved the way for enhanced medical applications, leading to the development of specialized versions such as MedSAM [77] and AutoSAM [78] for medical image segmentation, which have achieved performance levels comparable to U-Net. Recent advancements also include universal ultrasound foundation models, such as USFM [79], enhanced SAM-based models like MedSAM [82] and SAMUS [83]. Additionally, the vision–language foundation model EchoCLIP [82] has been developed for echocardiogram interpretation. Looking ahead, it is imperative to explore multimodal foundation models that integrate sensor, vision, and language inputs to address challenges related to data acquisition, analysis, and interpretation.

D. Explore Novel Application Areas for Ultrasound

DL-based ultrasound diagnostics present significant opportunities for exploring new clinical application areas. One emerging area of interest is nutritional assessment. Currently, measurement of body composition as an indicator of nutritional status is often derived from manual calculations [177]. While most automatic analysis has predominantly focused on CT data [178], [179], [180], the emergence of low-cost portable ultrasound devices offers new opportunities for ultrasound data acquisition and automated clinical evaluations [181]. Other underexplored applications include improving real-time physiological monitoring and remote patient management with wearable ultrasound devices [182], early detection of metabolic disorders to identify signs of conditions such as fatty liver disease [183], and personalized medicine [162].

E. Improve the Deployment of DL Models in Ultrasound Systems

Once DL models are deployed within an ultrasound system in clinical environments, particularly on real-time or resource-constrained ultrasound systems, overall performance may degrade. To improve model deployment, future work should emphasize the use of lightweight, edge-efficient DL models such as SqueezeNet [184] for portable systems, real-time

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monitoring of CPU/GPU load during inference, maintenance of manual override options for clinicians, and routine validation of model performance with new datasets to mitigate domain drift.

VII. Conclusion

In this review, we provide an overview of the rapidly evolving field of DL and its recent applications in ultrasound imaging. We summarize recent advancements in supervised learning, self-supervised learning, discriminative models, generative AI, foundation models and other miscellaneous models, and introduce their applications. Then, we provide a comprehensive survey on the state-of-the-art work of DL-based ultrasound image analysis across a range of anatomical sites and clinical tasks. After that, we summarize DL-based clinical ultrasound applications, and then highlight the potential of foundation models in clinical ultrasound. Finally, we outline persistent research challenges and key future directions.

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